Speech-to-Text (STT) Module Methodology

1. Module Overview

The Speech-to-Text (STT) module is responsible for converting visitor voice input into text that can be processed by the NLP pipeline. The module is built on top of the Groq Whisper API and extends it with two key mechanisms: prompt injection for domain-specific vocabulary guidance, and LLM-based post-correction to fix artifact name misspellings. The complete processing flow is shown in Figure 1.

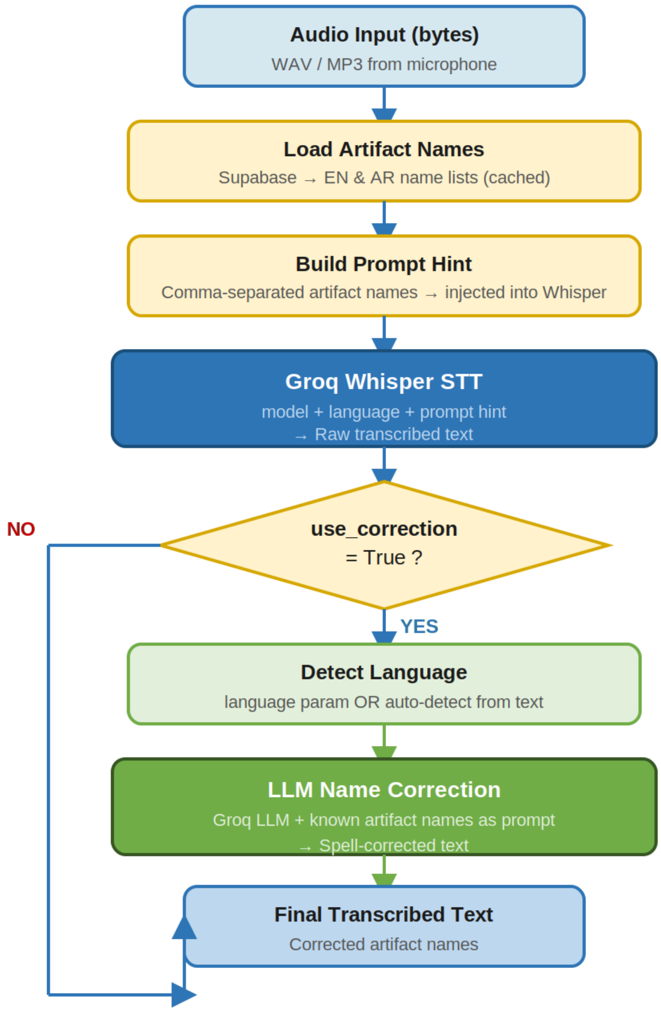

Figure 1: STT Module Full Processing Pipeline.

2. Why Prompt Injection Instead of Fine-Tuning

A core design decision in this module is the choice to use prompt injection rather than model fine-tuning to teach the Whisper model the names of museum artifacts. This decision was made for the following reasons:

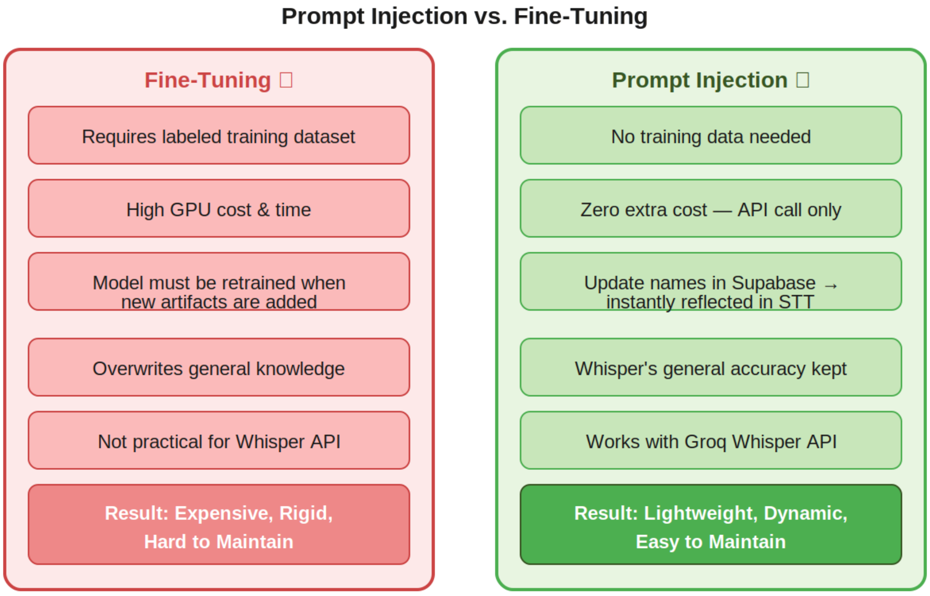

Figure 2: Prompt Injection vs. Fine-Tuning — Comparison.

2.1 The Problem: Domain-Specific Vocabulary

Speech recognition models like Whisper are trained on general-purpose audio datasets. When a visitor says the name of a rare ancient Egyptian artifact such as 'Narmer Palette' or an Arabic name like 'مسلة تحتمس,' the model may transcribe it incorrectly because these names are rare or absent from general training data.

Two possible solutions exist: (1) fine-tuning the model on domain-specific audio, or (2) injecting the known vocabulary as a prompt hint at inference time.

2.2 Why Fine-Tuning Was Rejected

Fine-tuning was evaluated and rejected for the following reasons:

- Requires a labeled audio dataset of visitors speaking artifact names — which does not exist and would be expensive to collect.

- Requires significant GPU compute time and cost to retrain, which is impractical for a museum deployment.
- Every time a new artifact is added to the museum collection, the model would need to be retrained from scratch.
- Fine-tuning risks degrading general transcription accuracy by overfitting to domain vocabulary.
- Groq's Whisper API does not expose a fine-tuning endpoint — it is an inference-only service.

2.3 How Prompt Injection Works

Whisper accepts an optional 'prompt' parameter that acts as a prior context hint. When provided, Whisper biases its transcription toward vocabulary present in the prompt. The system exploits this by injecting the names of all known artifacts as a comma-separated string:

```
kwargs["prompt"] = "Narmer Palette, Nefertiti Bust, Rosetta Stone, لوحة نارمر, تمثال نفرتيتي ..."
```

These names are loaded from Supabase at startup and cached in memory. The prompt is updated automatically whenever the Supabase artifact database is updated — no model retraining is required. The prompt is capped at 1,000 characters to stay within Whisper's token limit.

Aspect	Fine-Tuning	Prompt Injection (Chosen)
Training data needed	Yes — labeled audio	No
Compute cost	High (GPU training)	None (inference only)
Update when new artifact added	Retrain model	Update Supabase only
Whisper API compatible	No	Yes
Risk to general accuracy	High (overfitting)	None

Table 1: Fine-Tuning vs. Prompt Injection Comparison.

3. LLM Post-Correction Stage

Even with prompt injection, Whisper may still produce minor misspellings of artifact names — especially for rare Arabic names or less common transliterations. A second correction stage is applied using a Groq LLM after transcription.

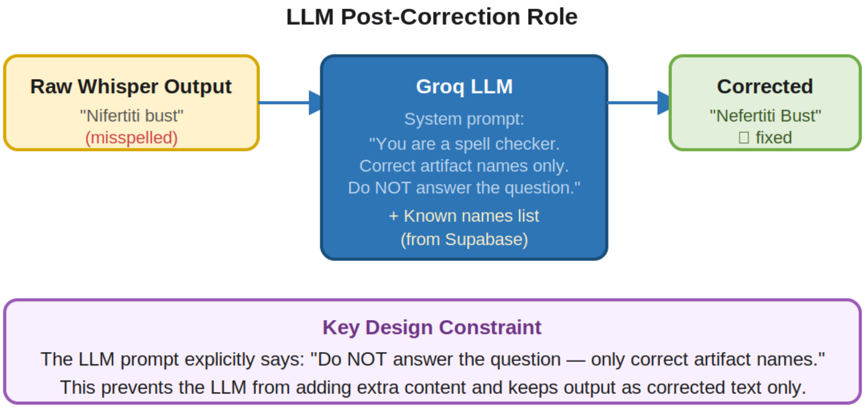

Figure 3: LLM Post-Correction Role and Design Constraint.

3.1 Design Constraint: LLM Must Not Answer the Question

The most critical design constraint in this stage is preventing the LLM from generating a conversational answer. Without an explicit instruction, an LLM receiving a visitor question like 'Tell me about the Narmer Palette' would produce a full answer rather than simply correcting the name spelling.

This is prevented by the system prompt, which explicitly states:

```
"You are a spell checker for a museum transcription system."
"Your ONLY job is to correct misspellings of ancient Egyptian and Greco-Roman artifact names."
"Do NOT answer the question. Do NOT add extra words. Just output the corrected text."
```

This constraint ensures that the LLM acts purely as a post-processor — its output is the same visitor sentence with artifact names corrected, not a response to the visitor's query.

3.2 Language-Aware Correction

The LLM correction is language-aware. If the transcription language is detected as Arabic ('ar'), the system injects the Arabic artifact name list into the LLM prompt. For English ('en'), the English names are injected. This ensures the model references the correct vocabulary for the detected language and avoids cross-language confusion.

3.3 Two-Layer Strategy Summary

The combination of both layers provides complementary coverage:

Layer	Mechanism	Handles
Layer 1 — Prompt Hint	Inject names into Whisper prompt	Biases STT toward correct vocabulary at transcription time
Layer 2 — LLM Correction	Groq LLM spell-check post-processing	Fixes residual misspellings after transcription

Table 2: Two-Layer Artifact Name Correction Strategy.

4. Module Configuration Parameters

Parameter	Value / Source	Description
stt_model	settings.stt_model	Groq Whisper model name
llm_model	settings.llm_model	Groq LLM model for correction
temperature	0.0 (both stages)	Deterministic output — no randomness
max_tokens (LLM)	200	Short output — corrected sentence only
prompt max length	1,000 chars	Whisper prompt token limit safety cap
use_correction	True (default)	Can be disabled to skip LLM stage
language	'ar' / 'en' / None	Controls which name list is injected

Table 3: STT Module Configuration Parameters.

3.X Theoretical Comparison of STT Approaches

To evaluate the suitability of different speech recognition approaches for the museum guide system, a theoretical comparison was conducted between Vosk, Whisper Large V3, and the proposed Whisper-based architecture. The comparison considers speech recognition capability, support for domain-specific vocabulary, scalability, computational requirements, and overall suitability for museum environments.

Table X. Theoretical Comparison of STT Approaches

Feature	Vosk	Whisper Large V3	Proposed Architecture
Speech Recognition Engine	Kaldi-based	Transformer-based	Whisper Large V3
Multilingual Support	Moderate	High	High
Noise Robustness	Moderate	High	High
Rare Artifact Name Recognition	Limited	Moderate	High
Vocabulary Adaptation	Requires retraining	Limited	Dynamic prompt injection
Post-Processing Correction	No	No	LLM-based correction
Update When New Artifact Added	Model update required	No adaptation mechanism	Update artifact database only
Computational Cost	Low	High	High
Scalability	Moderate	High	High
Suitability for Museum Environment	Moderate	Good	Excellent

The comparison demonstrates that the proposed architecture provides the most suitable balance between transcription accuracy and adaptability to museum-specific terminology. Therefore, it was selected as the speech recognition approach for the proposed system.